

Notes: Acemoglu (Economic Policy, 2025)

The Simple Macroeconomics of AI

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1 Big picture

- **Research question.** How large are the medium-run macroeconomic effects of recent AI advances, especially generative AI and related computer vision technologies? The paper asks whether claims of very large GDP, productivity, wage, and inequality effects are plausible over roughly a 10-year horizon.
- **Main idea.** A simple task-based model disciplines the macro estimates. If AI raises productivity by reducing the cost of performing existing tasks, then aggregate effects are pinned down by a version of Hulten’s theorem: the macro gain equals the economic share of tasks affected times the average task-level cost saving.
- **Core contrast.** Many public forecasts start from impressive micro examples and scale them up. Acemoglu instead asks how much of the economy those examples can plausibly affect within 10 years and how large the cost savings are likely to be after adjustment costs and task difficulty are considered.
- **Main finding.** The predicted TFP gains are positive but modest. The baseline upper bound is about 0.66% over 10 years, or roughly 0.064 percentage points of additional annual TFP growth. Once the distinction between easy and hard tasks is incorporated, the estimate falls to about 0.53% over 10 years.
- **GDP versus TFP.** GDP can rise by more than TFP if AI induces more capital accumulation. The paper estimates GDP effects of about 0.93–1.16% over 10 years under a modest investment response, and about 1.4–1.56% under a larger investment response.
- **Wages and inequality.** AI exposure is more evenly distributed across education and wage groups than earlier automation technologies. This means the wage effects are not as sharply anti-worker as robotics or older automation, but the paper finds little reason to expect AI to reduce inequality. It may slightly raise between-group inequality and widen the gap between capital and labour income.
- **New tasks matter.** The estimates focus mainly on automation and task complementarities in existing tasks. If AI creates genuinely useful new tasks for workers, the benefits could be larger and more labour-friendly. But if AI creates manipulative or socially harmful tasks, GDP can rise while welfare falls.
- **Economic takeaway.** AI is economically important, but the medium-run macro effects are likely to be much smaller than many headline forecasts unless AI development is redirected toward creating reliable tools and new productive tasks for workers.

2 Incremental contribution of the paper

- **A disciplined macro accounting framework.** The paper turns AI forecasting into an accounting problem: exposure share times cost savings, with careful attention to whether the relevant tasks will actually be affected within a decade.
- **Task-level foundation.** The paper uses the task-based framework of Acemoglu and Restrepo to distinguish automation, task complementarity, deepening of automation, and new tasks.

- **Clear use of Hulten’s theorem.** Rather than building an unconstrained forecast, the paper shows that task-level cost savings aggregate according to task shares in GDP. This puts a hard discipline on large macro claims.
- **Easy versus hard tasks.** The paper adds an important distinction between tasks that AI can learn easily and tasks where success depends on context, judgment, tacit knowledge, and hard-to-observe outcomes. This is one of the main reasons the revised estimates are smaller.
- **Quantitative synthesis of recent evidence.** The paper combines task-exposure estimates from Eloundou et al., feasibility estimates from Svanberg et al., and experimental productivity estimates from Peng et al., Noy and Zhang, and Brynjolfsson et al.
- **GDP-welfare distinction.** The paper emphasizes that measured GDP is not the same as welfare. Extra investment uses resources, and some AI-generated activities may be manipulative or defensive rather than welfare-enhancing.
- **Distributional analysis.** The paper adapts the general equilibrium wage and inequality framework of Acemoglu and Restrepo to AI-exposed occupations and demographic groups.
- **Policy relevance.** The paper argues that the largest social gains are unlikely to come automatically from current industry priorities. More useful outcomes may require technologies and institutions that create new tasks and improve worker productivity directly.

3 Conceptual framework

3.1 Task-based production

The paper uses a task-based production model. Output is produced by a continuum of tasks. Each task can be performed by labour or capital, and the allocation of tasks across labour and capital determines productivity, wages, factor shares, and the effect of new technologies.

- **Tasks.** Production requires many tasks indexed by z .
- **Labour and capital.** Labour and capital have different productivities across tasks.
- **Automation.** Automation expands the set of tasks performed by capital rather than labour.
- **Task complementarity.** AI may help workers perform remaining tasks more effectively without fully displacing them.
- **New tasks.** AI may create tasks that were not previously part of production.

Interpretation. AI is not treated as an abstract productivity shock. The model asks where AI enters the production process and whether it substitutes for labour, complements labour, deepens existing automation, or creates new activities.

3.2 Four channels through which AI can matter

The paper emphasizes four possible channels:

- **Extensive-margin automation.** AI takes over tasks previously performed by workers. Examples include text summarization, classification, some clerical tasks, pattern recognition, and computer vision tasks.
- **Task complementarity.** AI helps workers perform existing tasks better or faster. It may automate subtasks while allowing workers to specialize in the remaining components of a job.
- **Deepening of automation.** AI improves tasks that were already automated, such as IT security or automated inventory control.
- **New tasks.** AI creates new products or new labour-intensive tasks. These can be socially valuable, but some can also be manipulative or defensive.

The quantitative analysis focuses mainly on the first two channels: automation and task complementarity in existing tasks.

3.3 Hulten's theorem and aggregate productivity

The core formula is a version of Hulten's theorem. For small productivity improvements in existing tasks,

$$d \ln TFP = \int_0^N \chi(z) \pi_L(z) dz,$$

where $\chi(z)$ is the GDP share of task z and $\pi_L(z)$ is the task-level productivity improvement or cost saving caused by AI.

If the average cost saving among impacted tasks is $\bar{\pi}$, then the expression becomes approximately:

$$d \ln TFP = \bar{\pi} \times \text{GDP share of tasks impacted by AI.}$$

Interpretation. Very large aggregate effects require either a very large share of tasks to be affected, very large cost savings in those tasks, or both. Impressive micro evidence is not enough if the affected tasks are a small share of GDP.

3.4 From TFP to GDP

GDP effects are larger than TFP effects if the capital stock responds:

$$d \ln Y = d \ln TFP + s_K d \ln K,$$

where s_K is the capital share. As a simple benchmark, if the capital stock rises proportionately with TFP, then:

$$\text{GDP gain} = \frac{\text{TFP gain}}{1 - s_K}.$$

Using a capital share of 0.43, the multiplier is about $1/(1 - 0.43) \approx 1.75$.

Interpretation. GDP can increase more than TFP because the economy invests more capital, but this does not mechanically mean welfare rises by the same amount. The extra investment uses resources that otherwise could have been consumed.

3.5 Wage effects

Automation has two opposing effects on wages:

- **Productivity effect.** AI lowers production costs and raises output, which tends to increase labour demand.
- **Displacement effect.** AI removes workers from tasks they used to perform, which tends to reduce labour demand for the affected workers.

Task complementarity can also have ambiguous wage effects. Even when workers become physically more productive in a task, equilibrium task prices can fall. With a low elasticity of substitution across tasks, the wage gains from labour-augmenting improvements can be limited.

3.6 Easy versus hard tasks

A key distinction is between easy-to-learn and hard-to-learn tasks.

- **Easy-to-learn tasks** have reliable observable outcome measures and a relatively simple mapping from actions to outcomes. Examples include classification, transcribing, standard writing, computing, grading, and some coding subtasks.
- **Hard-to-learn tasks** depend heavily on context, judgment, tacit information, and outcomes that are hard to observe or evaluate. Examples include diagnosis, advising, care, hiring, instruction, representation, and many complex decision-making activities.
- **Why this matters.** Existing proof-of-concept studies mostly cover relatively easy tasks. Extrapolating their cost savings to hard tasks likely overstates the macro effect.

Interpretation. AI can learn from data and human behavior, but when the objective function is unclear or success is context-dependent, the model may reproduce average human behavior rather than expert performance. This limits near-term productivity gains.

4 Quantitative inputs and estimates

4.1 Exposure of tasks to AI

The paper starts with the task-exposure estimates of Eloundou et al. These estimates are based on O*NET task and detailed work activity descriptions and GPT-based classification of exposure to LLM and computer vision technologies.

- The paper uses the granular automation/exposure index rather than a purely binary exposure measure.
- Tasks for which 50% or less of activities are assessed as impacted are set to zero.
- The exposed tasks are aggregated to occupations and then weighted by wage bills.
- The result is that about 20% of US labour tasks are exposed to AI in the broad sense.

Next step. Exposure is not the same as profitable adoption. The paper uses Svanberg et al.’s estimate that 23% of exposed computer-vision tasks can be feasibly and profitably automated within 10 years. Applying this feasibility share to the 20% exposure estimate gives:

$$0.23 \times 0.20 = 0.046.$$

Thus, the baseline GDP share of tasks impacted by AI within 10 years is 4.6%.

4.2 Cost savings from AI

The paper uses experimental studies of AI productivity gains to estimate labour cost savings.

- **Noy and Zhang.** Generative AI improves performance in writing-related tasks, with gains especially concentrated among initially lower-performing workers.
- **Brynjolfsson et al.** A generative AI tool in customer service raises speed by about 14% on average, with gains concentrated among lower-performing workers.
- **Peng et al.** GitHub Copilot substantially speeds up a specific coding task, but the setting is narrow and very well matched to the tool.

The baseline averages Noy and Zhang with Brynjolfsson et al. and uses a 27% labour cost saving. Because labour is only part of total cost, this is multiplied by the labour share of exposed industries, estimated at 0.535:

$$0.27 \times 0.535 = 0.144.$$

Thus, the baseline overall cost saving in impacted tasks is 14.4%.

4.3 Baseline TFP estimate

Combining the 4.6% impacted GDP share and the 14.4% overall cost saving gives:

$$\text{TFP gain over 10 years} = 0.046 \times 0.144 \approx 0.0066.$$

- This equals about 0.66% higher TFP after 10 years.
- It corresponds to about 0.064 percentage points of additional annual TFP growth.
- Including the larger Peng et al. productivity gains raises the estimate to roughly 0.89%, still modest relative to many public forecasts.

Interpretation. The baseline result is positive and economically meaningful, but far below claims that AI will add multiple percentage points to annual GDP growth over the next decade.

4.4 Refining the estimate with easy and hard tasks

The paper then separates exposed tasks into easy and hard categories. The classification uses verbs, intermediate work activities, topic modelling, and a gradient-boosted tree trained on manually classified tasks.

- About 72.6% of wage-bill-weighted exposed tasks are classified as easy.

- Easy exposed tasks account for about 3.3% of GDP.
- Hard exposed tasks account for the remaining 1.3% of GDP.
- Easy tasks keep the 27% labour cost saving, which becomes 14.4% after multiplying by the labour share.
- Hard tasks are assigned a 7% labour cost saving, which becomes 3.7% after multiplying by the labour share.

The revised calculation is:

$$0.033 \times 0.144 + (0.046 - 0.033) \times 0.037 \approx 0.0053.$$

Result. The refined estimate is about 0.53% TFP growth over 10 years.

4.5 GDP estimates

Using the proportional-capital benchmark, the paper converts TFP to GDP by multiplying by about 1.75:

- Baseline TFP gain of 0.66% implies GDP gain of about 1.16%.
- Easy/hard-adjusted TFP gain of 0.53% implies GDP gain of about 0.93%.
- Using a larger investment response from the Acemoglu-Restrepo framework gives upper-bound GDP effects of about 1.56% without the easy/hard distinction and 1.4% with it.

Interpretation. GDP effects are larger than TFP effects, but this partly reflects additional capital accumulation, not a pure welfare gain.

4.6 New bad tasks and welfare

The paper also discusses AI-created tasks that may have negative social value. Examples include deepfakes, manipulative digital advertising, addictive social media, and cyberattack/cyberdefense arms races.

- New tasks can raise measured GDP if they generate revenue.
- But some new tasks may reduce welfare if they manipulate users, waste attention, or force defensive spending.
- Using numbers from Bursztyn et al. on social media welfare, the paper illustrates that AI-related bad tasks could appear to add about 2% to GDP while reducing welfare by about 0.72% in consumption-equivalent units.

Interpretation. GDP is not sufficient for evaluating AI. Some AI-generated economic activity can be privately profitable but socially harmful.

5 Identification and interpretation

5.1 What the paper can identify

The paper does not run a single causal experiment on aggregate AI. Instead, it gives a disciplined quantitative forecast based on a model, existing task-exposure measures, experimental productivity estimates, and transparent assumptions.

- The paper identifies how large AI’s medium-run macro effects are likely to be if current estimates of exposure, feasibility, and cost savings are approximately correct.
- It identifies why large micro effects do not necessarily imply large macro effects.
- It identifies the assumptions needed for bigger aggregate effects: more tasks affected, greater cost savings, faster adoption, or many valuable new tasks.

5.2 Why the estimates are upper bounds

Several forces suggest that even the 0.53–0.66% TFP estimates may be upper bounds.

- AI investment and adoption are still limited outside large firms and a few sectors.
- Organizational change is slow, and digital technologies often have J-shaped productivity effects with delayed gains.
- Current experimental evidence comes mainly from easy-to-learn tasks.
- Some exposed tasks may not be profitable to automate or complement within 10 years.
- Generative AI tools may be concentrated in a small number of firms, which could slow diffusion to small and medium enterprises.

5.3 Why productivity and wages differ

The paper emphasizes that productivity gains do not mechanically translate into wage gains.

- Automation can increase output while displacing labour from tasks.
- Task complementarity can raise physical productivity but reduce task prices.
- Extra capital accumulation can raise GDP more than average wages.
- AI may widen the capital-labour income gap even if no education group suffers very large wage losses.

5.4 Why low-skill productivity gains can raise inequality

A striking theoretical point is that improving low-skill productivity in some tasks does not guarantee lower inequality. If AI raises low-skill productivity in a subset of tasks without creating new tasks for low-skill workers, the equilibrium reallocation of labour can reduce low-skill wages elsewhere or strengthen competition against low-skill workers in remaining tasks.

Interpretation. Distributional effects require a general equilibrium analysis. It is not enough to observe that AI helps lower-performing workers in one experimental setting.

6 Tables and figures

6.1 Figure 1 and Table 1: Exposure and wage effects

Read Figure 1:

- Figure 1 plots AI exposure across 500 demographic groups against average hourly wages.
- Marker size reflects employment, and shading indicates education group.
- AI exposure is more evenly spread across the wage distribution than previous automation exposure.
- Less-than-high-school groups have relatively low exposure, while some-college and college groups tend to have relatively high exposure.

Economic message: AI exposure is not concentrated as narrowly among low-education workers as robotics and earlier automation. This is why the predicted inequality effects are less extreme than for earlier automation waves.

Read Table 1:

- Table 1 summarizes AI exposure and wage effects by education group.
- Baseline exposure is lowest for workers with less than high school and highest for workers with a Bachelor's degree.
- Direct displacement effects are negative for all education groups.
- Full equilibrium wage effects are small and positive for all education groups because productivity and reallocation effects partly offset displacement.
- The easy/hard-adjusted estimates imply a GDP effect of 1.4% and an average-worker wage effect of 0.77% over 10 years.

Economic message: The model predicts modest wage gains on average, no dramatic education-group wage collapse, and a larger GDP effect than wage effect, implying a higher capital share.

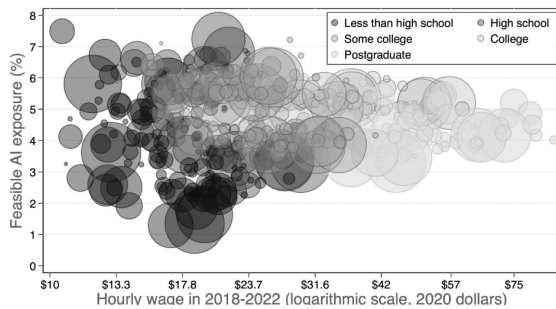


Figure 1: Distribution of AI exposure across the wage distribution.

Table 1. Exposure and wage effects by education groups

	Baseline exposure			Exposure adjusted for easy and hard task				
	(1) Baseline AI exposure	(2) Direct effect of AI exposure	(3) Total wage effect of AI exposure	(4) Exposure to easy tasks	(5) Exposure to hard tasks	(6) Direct effect of exposure to easy and hard tasks	(7) Total wage effect of exposure to easy and hard tasks	(8) Direct task displacement 1990-2016
Workers with less than high school degree	0.0318	-0.0323	0.0157	0.0235	0.0003	-0.0356	0.0132	0.2690
Workers with high school degree	0.0464	-0.0617	0.0106	0.0337	0.0128	-0.0649	0.0079	0.2706
Workers with some college	0.0494	-0.0636	0.0139	0.0337	0.0138	-0.0709	0.0112	0.1886
Workers with Bachelor's degree	0.0323	-0.0733	0.0060	0.0382	0.0140	-0.0765	0.0040	0.0684
Workers with postgraduate degree	0.0405	-0.0498	0.0098	0.0299	0.0106	-0.0530	0.0081	0.0343
Average worker	0.0462	-0.0612	0.0101	0.0336	0.0128	-0.0644	0.0077	0.2107
GDP			0.0156				0.0140	

Table 1: Exposure and wage effects by education group.

6.2 Figures 2 and 3: Decomposition and demographic wage effects

Read Figure 2:

- The first panel shows the direct productivity effect, which is nearly uniform across groups.
- Adding industry shifts barely changes the distribution.
- Adding direct task displacement creates more variation, especially among lower-wage groups.
- Adding ripple effects partially reshuffles the pattern but does not create the severe inequality effects associated with earlier automation.

Read Figure 3:

- Figure 3 breaks wage effects down by gender, nativity/race group, and education.
- Men generally see positive real wage effects across education groups.
- Women, especially low-education women, see weaker gains or small losses.
- The largest concern is for low-education women, especially native-born white women in the plotted categories.

Economic message: AI's wage effects are not uniformly progressive. Even with broad exposure and small average effects, some demographic groups can be hurt in equilibrium.

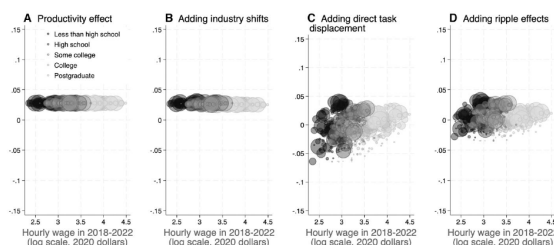


Figure 2: Decomposition of productivity, industry-shift, direct displacement, and ripple effects.

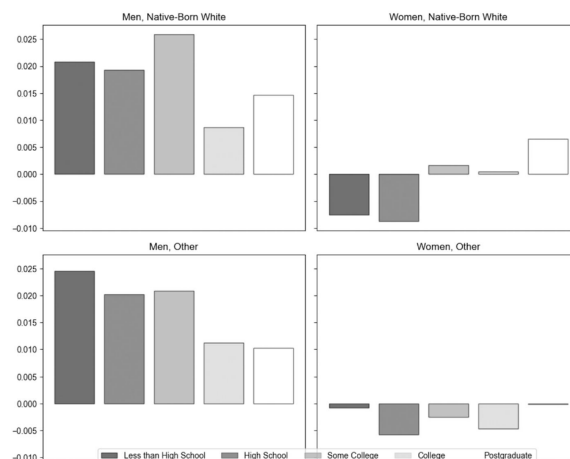


Figure 3: Total wage effect of AI exposure by gender and demographic group.

7 Short summary

- The paper evaluates medium-run AI effects using a task-based macro framework.
- The main aggregate productivity formula is Hulten-style: task share times task-level cost saving.
- Eloundou et al.'s exposure estimates imply that about 20% of labour tasks are exposed to AI.

- Svanberg et al.’s feasibility estimate implies that about 23% of exposed tasks are likely to be profitably affected within 10 years.
- Combining these gives an impacted GDP share of about 4.6%.
- Baseline labour cost savings are 27%, and industry labour shares convert this to 14.4% total cost savings.
- The baseline TFP effect is therefore about 0.66% over 10 years.
- After separating easy and hard tasks, the estimate falls to about 0.53%.
- GDP effects are larger, about 0.93–1.16% under a modest investment response and about 1.4–1.56% under a stronger one.
- AI exposure is more evenly distributed across demographic groups than previous automation, so the wage effects are not strongly concentrated on low-education workers.
- The model still predicts no clear reduction in inequality and a likely widening of the capital-labour income gap.
- AI-created new tasks could be very beneficial if they raise worker productivity, but some AI-created tasks may be socially harmful and can make GDP misleading.

8 Conclusion

8.1 Core conclusion

Acemoglu concludes that the medium-run macroeconomic effects of recent AI advances are likely to be positive but modest. The paper does not deny that AI is important. Instead, it argues that the size of the 10-year aggregate effect is constrained by the limited share of tasks likely to be profitably affected and by the moderate cost savings observed so far.

8.2 Economic intuition

The central intuition is simple: even powerful tools have limited macro impact if they affect a small part of the economy or if their cost savings are modest. A large language model can dramatically improve some writing, coding, or customer-service tasks, but the macroeconomy contains many tasks that are hard to learn, hard to evaluate, or slow to reorganize around.

8.3 Incremental contribution revisited

The paper’s main contribution is not a new empirical estimate from a single dataset. It is a disciplined framework for translating micro AI evidence into macro forecasts. It shows why exposure, feasibility, task difficulty, capital accumulation, and welfare all need to be separated.

8.4 Implications

The results imply that policymakers and researchers should be cautious about treating AI as an automatic source of explosive productivity growth. The most socially useful path may involve

developing AI tools that create new tasks for workers, provide reliable context-specific information, and raise the productivity of teachers, nurses, repair workers, electricians, clerical workers, and other middle- and lower-paid workers.

8.5 Limitations

The estimates rely on speculative assumptions because AI diffusion is still early. The exposed-task share, feasibility share, task-level cost savings, easy-versus-hard classification, and investment response are all uncertain. The analysis also leaves out potential large breakthroughs in science and invention, as well as the possibility of valuable new tasks that are hard to quantify. The paper is therefore best read as a disciplined benchmark, not a final forecast.

8.6 Takeaway

The enduring takeaway is that AI's macroeconomic effects depend less on spectacular demonstrations and more on mundane aggregation: which tasks are affected, how large the cost savings are, how quickly firms adopt the tools, whether the tasks are easy or hard to learn, and whether AI creates useful new work for humans. Under current evidence, the 10-year productivity gains are meaningful but modest.